TUTORIAL

A first QC/MM system

> Updated for EasyHybrid 3.2.0 (2026-10-26). The figures and numbers come from a run of this tutorial with the 3.2.0 development version.

A hybrid QC/MM model describes a small region with a quantum chemical (QC) method and the rest of the system with a force field (MM). The QC region is where electrons matter: bonds that break, charge transfer, electronic excitation, orbitals. Everything else (the protein, the solvent) is treated by molecular mechanics at a fraction of the cost.

In this tutorial you will:

1. choose the QC region: the side chain of Tyr2 of chignolin;
2. set its QC method (AM1) and see the link atom that closes the cut bond;
3. optimize the QC/MM system;
4. compute the molecular orbitals of the QC region and draw the HOMO and the LUMO.

The whole tutorial takes about one minute. The system is `examples/tutorials/06_qcmm/1UAO_OPLS.pkl` in the EasyHybrid folder: chignolin (PDB 1UAO), a 10-residue β-hairpin, 138 atoms, OPLS-AA.

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1. Open the system

File → Open and choose `examples/tutorials/06_qcmm/1UAO_OPLS.pkl` (set the file type to *pkl*).

2. Select the QC region

The QC region should contain the chemistry of interest and be cut, if it must be cut, at a non-polar single bond far from it. For a tyrosine side chain the natural cut is the CA–CB bond: the QC region is the side chain, from CB to the hydroxyl hydrogen (15 atoms).

Select it with the **EasyHybrid Terminal** (**View → EasyHybrid Terminal**):

```
select obj=0 resi=2 not_name=N,H,CA,HA,C,O
```

The terminal answers *15 atoms* (Figure 1). Selections made in the 3D view (Viewing mode) work the same way.

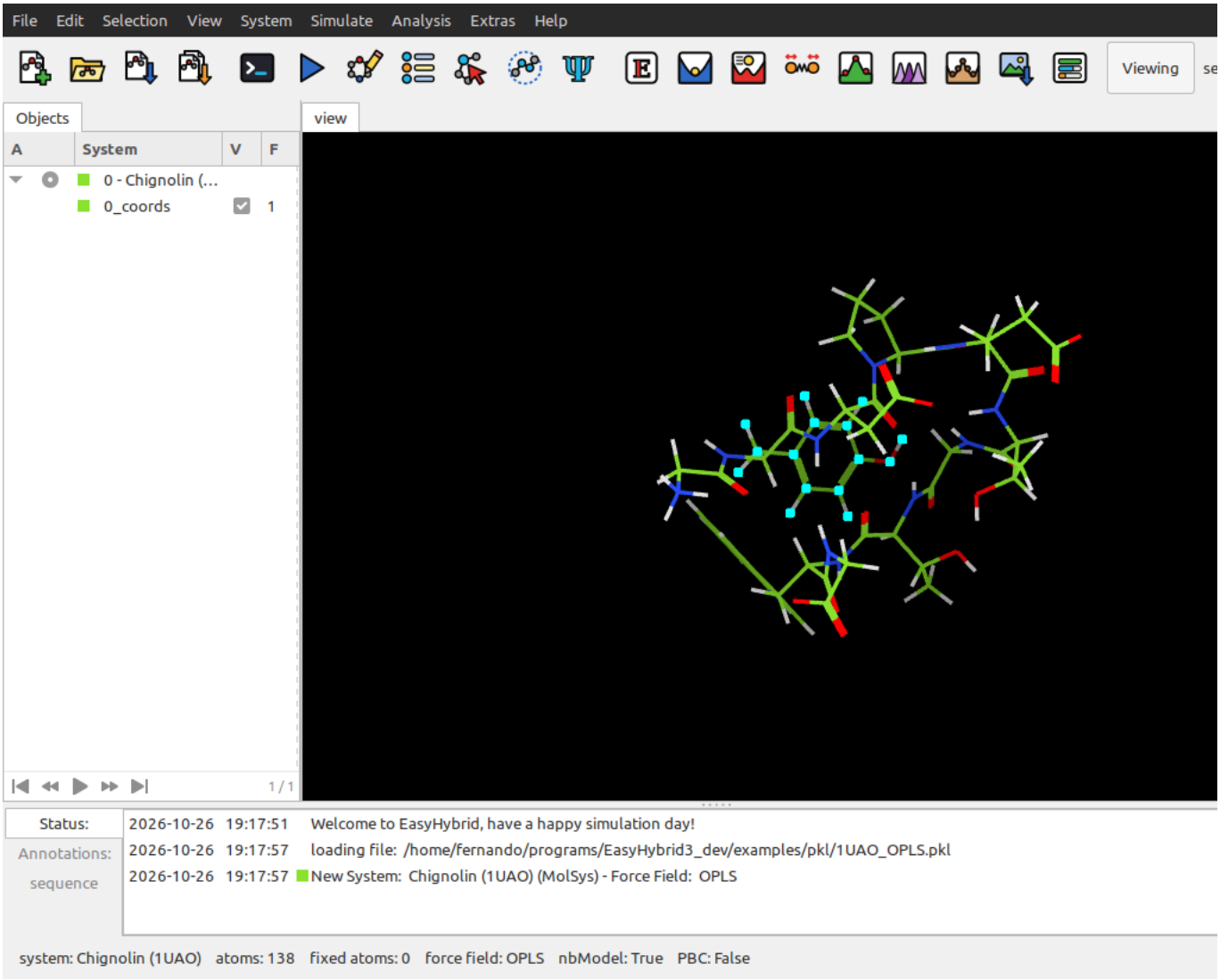


Figure 1. The 15 atoms of the Tyr2 side chain selected.

3. Define the QC region and its method

Selection → Set as QC Atoms. EasyHybrid asks *A new quantum region has been defined. Would you like to set up your QC parameters now?* Answer **Yes**: the QC model window opens (Figure 2).

Field	Value
Number of Atoms	15
QC Engine	pDynamo
Method	**am1**
Charge / Multiplicity	0 / 1 (a neutral phenol)

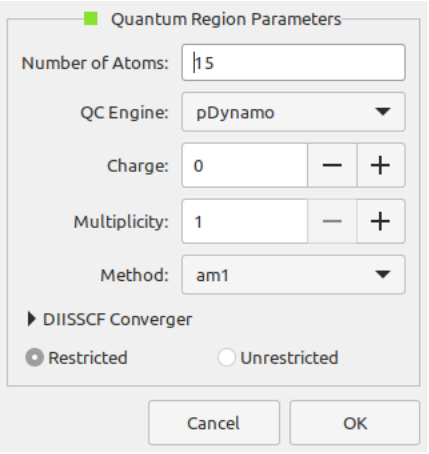


Figure 2. AM1 for the side chain.

Click **OK**. The QC atoms are drawn as balls and sticks (Figure 3) and the status bar shows `hamiltonian: am1`, `QC atoms: 15`.

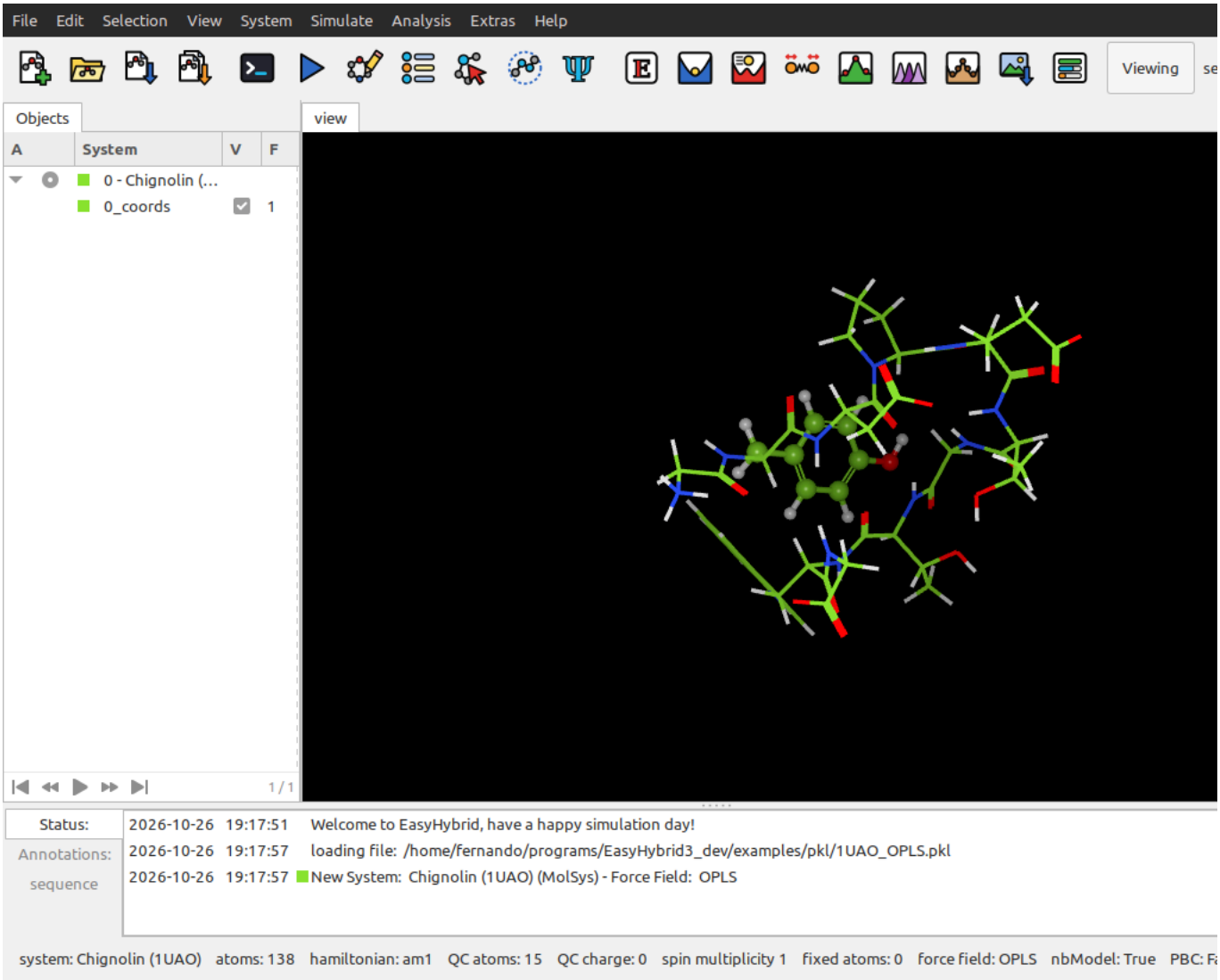


Figure 3. The QC region (balls and sticks) inside the MM peptide.

What happened at the boundary. The CA–CB bond crosses the boundary. pDynamo keeps CA in the MM region and puts a **link atom**, a hydrogen, on the QC side along the CA–CB bond, so that the QC region is a closed-shell molecule (here, 4-methylphenol). The QC atoms lose their MM charges, and the charges of the MM atoms of the residue are adjusted so that the MM part keeps an integer charge. The other MM atoms polarize the QC electrons through their charges (electrostatic embedding).

4. Optimize

Simulate → Geometry Optimization: Conjugate Gradient, 2000 iterations, RMS gradient 0.1, and **Run**. Each step computes the AM1 energy of the side chain in the field of the peptide plus the OPLS-AA energy of the rest: the optimization takes a few seconds.

5. The frontier orbitals

Analysis → Surfaces opens the **Orbitals and Surfaces** window. Choose the optimized object in **Object**: for a small QC region with a semiempirical method the orbitals are computed at once (*AM1, 40 basis functions*). The window shows (Figure 4):

- the **energy-level diagram** (occupied levels in blue, virtual in orange), with the HOMO–LUMO gap;
- the list of **orbitals** with their energies and main atomic contributions;
- for the chosen orbital, the contribution of each atom (bar plot) and, in **Composition (LCAO)**, the coefficient of every atomic orbital.

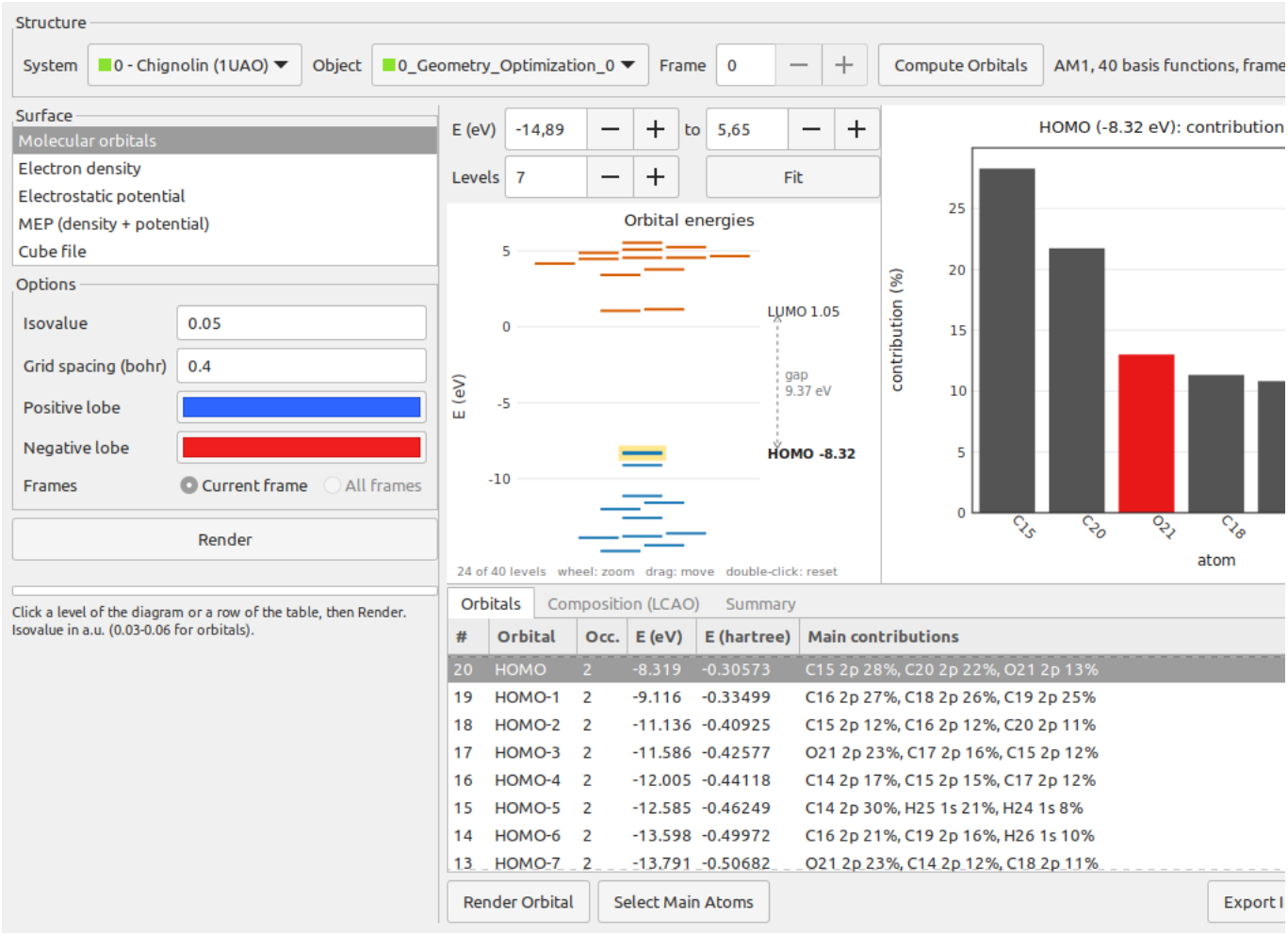


Figure 4. The orbitals of the QC region; the HOMO is selected.

	E (eV)	Main contributions
HOMO	−8.32	CG 27 %, CZ 22 %, OH 13 %, CE1 12 % (2p)
LUMO	+1.05	CD1 32 %, CE2 30 %, CD2 16 %, CE1 15 % (2p)
Gap	9.37	

The HOMO is the π orbital of phenol, with large coefficients on the ring carbons para to each other (CG, CZ) and on the oxygen lone pair: the electrons that tyrosine gives up when it is oxidized. The LUMO is a π* orbital of the ring. The atoms are numbered as in the system (CG is atom 15); the link hydrogen, listed as `HL11`, contributes almost nothing.

Select the HOMO and click **Render** (isovalue 0.05, grid 0.4 bohr): the orbital is added to the 3D view as a surface object under the system, with its two phases in blue and red (Figure 5). Do the same for the LUMO (Figure 6).

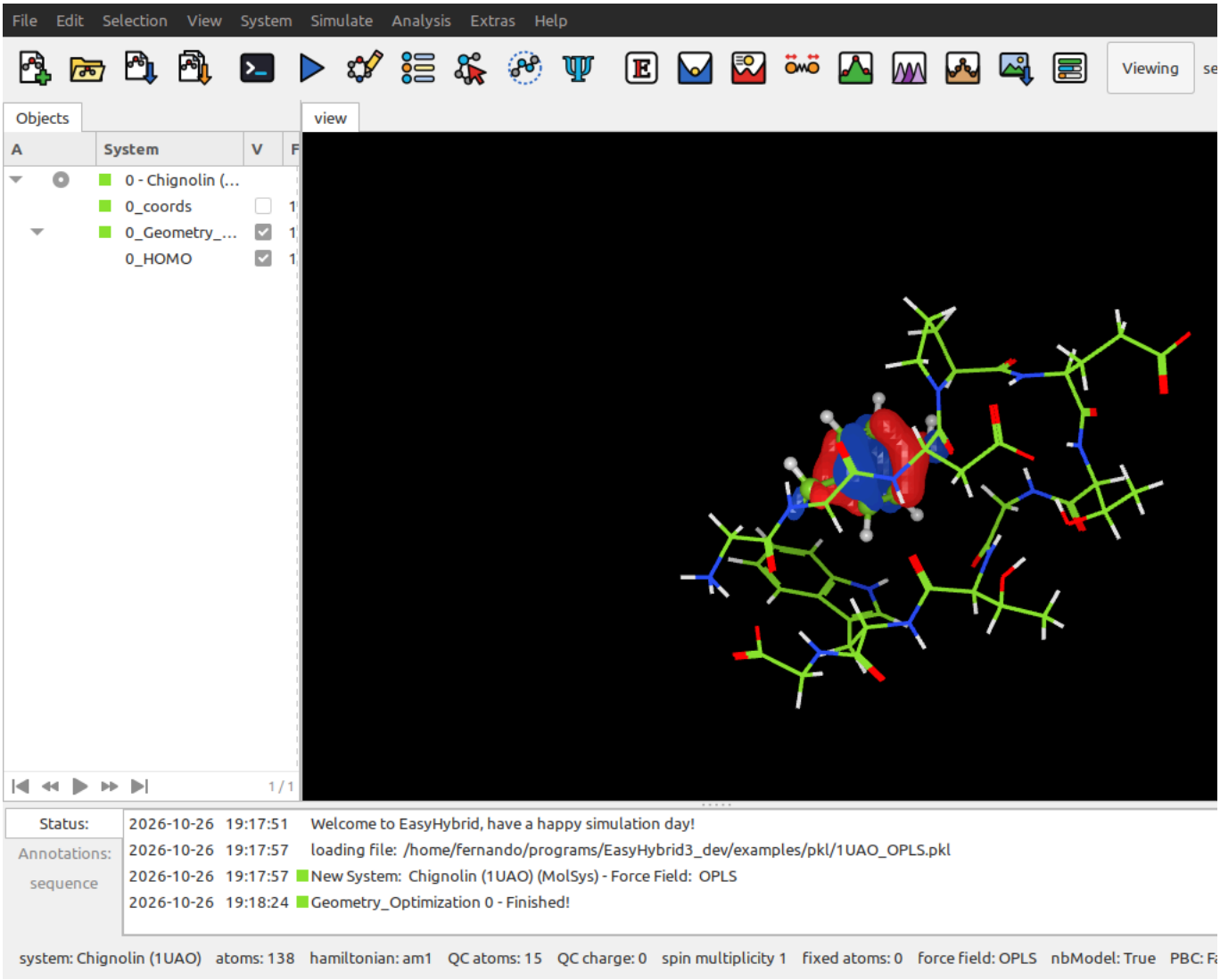


Figure 5. The HOMO of the Tyr2 side chain.

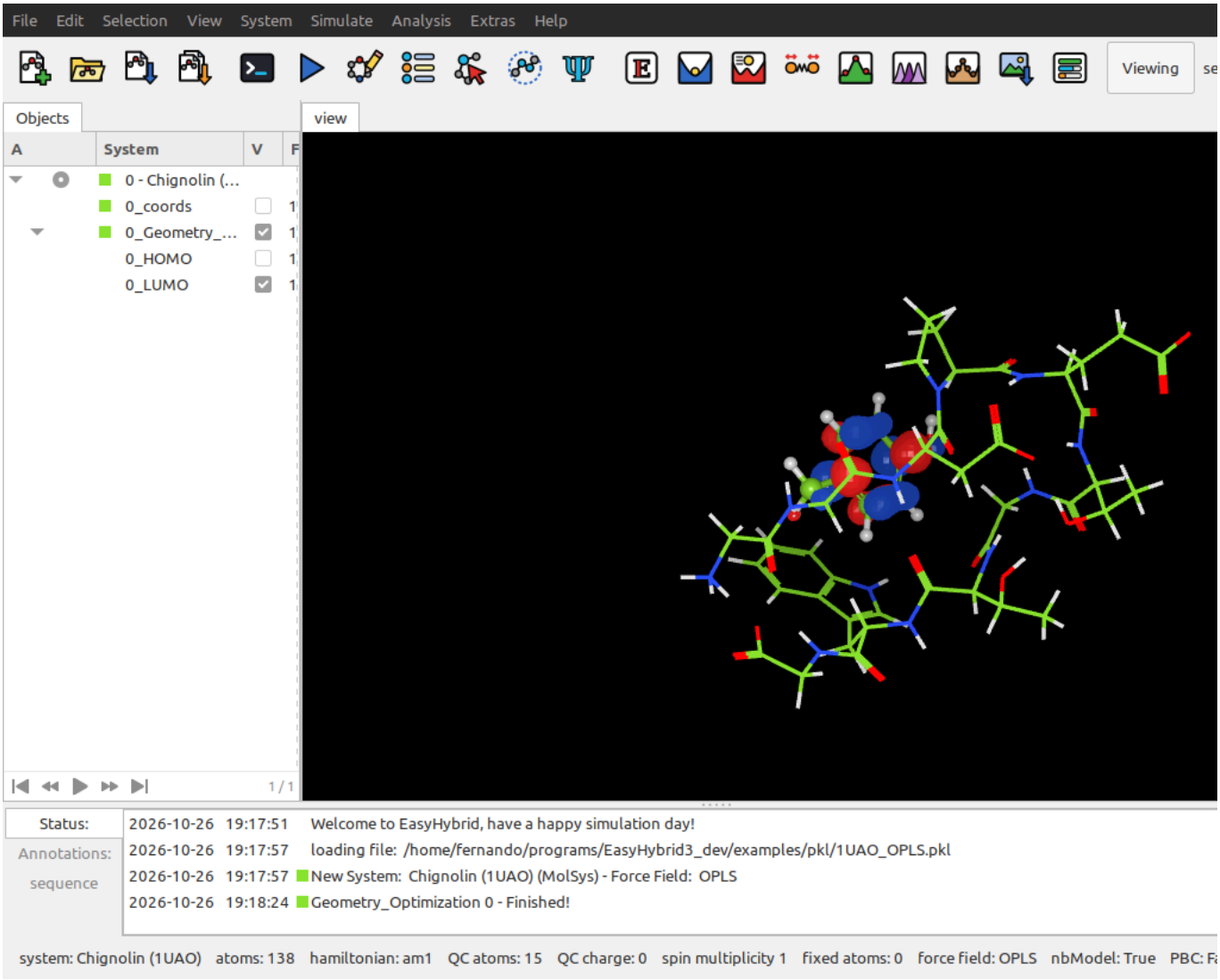


Figure 6. The LUMO.

Notes

- **Where to cut.** Never cut a polar or multiple bond (C–N, C=O, the peptide bond): the link atom would replace a group with a very different electronegativity. CA–CB is the usual cut for side chains; for a whole residue, cut the backbone at CA–C and N–CA of the neighbours.
- **Charge of the QC region.** It is the formal charge of what is inside, link atoms included: 0 for this phenol, −1 for a carboxylate (Asp, Glu), +1 for a protonated amine (Lys).
- **Orbitals with other methods.** pDynamo's semiempirical and DFT/HF models give orbitals directly. For xTB, ORCA or DFTB+ write a cube file with the program and use **Cube file** in the same window.
- Remove the QC region with **System → Remove → QC Atoms**.

EasyHybrid 3.2.0 · built 2026-10-29 from `dev` @ `3539da4` · https://github.com/ferbachega/EasyHybrid3